

AI-Based Ecological Predictive Cruise Control for Connected Electric Vehicles Using LSTM-Enhanced PID and Model Predictive Control

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Abstract - This paper presents an AI-based ecological predictive cruise control (Eco-PCC) system for connected electric vehicles (EVs) leveraging a stacked long short-term memory (LSTM) neural network integrated with two advanced control architectures: an LSTM-enhanced proportional–integral–derivative (LSTM-PID) controller with deceleration-anticipation feedforward, and an LSTM-driven receding-horizon model predictive controller (LSTM-MPC). The LSTM model predicts a 20-step velocity trajectory from a 100-step historical window, trained on a multi-scenario synthetic traffic dataset encompassing highway, urban, smooth, and variable driving profiles. Three controllers are benchmarked - a bang-bang baseline, LSTM-PID, and LSTM-MPC - across energy consumption, battery state-of-charge (SOC), speed tracking RMSE, and torque smoothness. Simulation results confirm that both LSTM-enhanced controllers achieve approximately 13% energy savings over the reactive bang-bang baseline (219.1 Wh), with LSTM-PID reaching 190.0 Wh (+13.3%) and LSTM-MPC reaching 190.7 Wh (+13.0%). Notably, LSTM-PID produces the smoothest torque profile (7.34 Nm vs 30.57 Nm for bang-bang), making it the preferred candidate for embedded deployment, while LSTM-MPC offers comparable energy savings with the added benefit of explicit horizon-based optimization.

Keywords – Electric Vehicle; Cruise Control; LSTM; Model Predictive Control; Regenerative Braking; Energy Efficiency; Autonomous Driving; PID Control.

I. INTRODUCTION

Range anxiety and energy efficiency remain primary adoption barriers for battery electric vehicles (EVs). Cruise control systems that blindly maintain a set speed waste energy through unnecessary acceleration – deceleration cycles, particularly in connected vehicle scenarios where real-time traffic information is available. Intelligent cruise control algorithms that anticipate upcoming speed changes and plan torque proactively can substantially reduce traction energy expenditure and improve regenerative braking capture.

Classical control approaches - proportional–integral–derivative (PID) and bang-bang controllers - react to speed error after it has accumulated, resulting in jerky torque profiles and missed regen opportunities [1],[2]. Model predictive control (MPC) mitigates this by planning a torque sequence over a finite horizon, but traditionally relies on hand-crafted traffic models that cannot adapt to real driving variability [3]. Recent advances in deep learning, particularly sequence-to-sequence LSTM networks, offer a data-driven path to traffic speed prediction [4],[5]. Integrating LSTM forecasts into cruise control creates a two-layer architecture: the outer LSTM layer anticipates traffic, while the inner control layer optimizes EV torque against the predicted reference. This hybrid paradigm has shown promise in simulation but lacks systematic three-way comparison studies spanning feedforward PID and full MPC variants in a physics-faithful EV model.

This paper makes the following contributions:

- (i) A stacked LSTM predictor trained with proper train/val split and no data leakage, capable of predicting 20-step velocity sequences.
- (ii) An LSTM-PID controller with a deceleration-only feedforward term that pre-emptively coasts ahead of speed drops without wasting energy on pre-emptive acceleration.

(iii) An LSTM-MPC controller with a normalized power cost function that prevents energy-weight dominance across varying operating conditions.

(iv) A full simulation study on a physics-based EV model including aerodynamic drag, rolling resistance, and regenerative braking.

The remainder of the paper is organized as: Section II presents the EV model and LSTM architecture; Section III details the three controllers; Section IV covers simulation results; Section V concludes the work.

II. SYSTEM MODEL

A. Electric Vehicle Dynamics

The EV is modelled as a single rigid body subject to four longitudinal forces at each timestep t with time interval $\Delta t = 0.1s$:

$$F_{net} = F_{prop} - F_{drag} - F_{roll} - F_{grav}$$

Where the propulsive force $F_{prop} = \tau/r_w$ is derived from motor torque τ and wheel radius r_w . Aerodynamic drag, rolling resistance, and gravity (on graded road) follow standard expressions. Vehicle acceleration $a = F_{net}/m$ updates velocity $v(t+1) = v(t) + a\Delta t$. Key parameters are: mass $m = 2500$ kg, $C_d = 0.28$, frontal area $A_f = 2.5$ m², $r_w = 0.36$ m, battery capacity $E_B = 48$ kWh.

Battery state-of-charge (SOC) is updated as:

$$SOC(t+1) = SOC(t) - P_{batt}\Delta t / (3600 \cdot E_B)$$

where $P_{batt} = P_{mech}/\eta_{motor}$ for motoring and $P_{batt} = P_{mech} \cdot \eta_{regen}$ for regeneration, with $\eta_{motor} = 0.92$ and $\eta_{regen} = 0.75$.

B. Traffic Dataset Generation

A synthetic dataset of 10,000 velocity sequences is generated by cycling through four driving scenarios constructed as closed-form time series with Gaussian noise and Gaussian smoothing ($\sigma = 2$):

$$V_{hw} = 25 + 8 \sin\left(\frac{t}{15}\right) + 3 \sin\left(\frac{t}{5}\right) + \varepsilon_1$$

$$V_{urb} = 15 + 12 \sin\left(\frac{t}{8}\right) \cdot \sin\left(\frac{t}{25}\right) + \varepsilon_2$$

where $\varepsilon \sim \mathcal{N}(0, \sigma_i^2)$. Smooth and variable profiles follow analogous forms. Velocities are clipped to $[0, 35]$ m/s. Input windows of length T in = 100 steps are paired with prediction targets of T out = 20 steps. Scalars are fitted exclusively on the training split (80%) to prevent data leakage.

C. LSTM Speed Predictor

The LSTM model (Fig. 1) takes a scaled velocity history and outputs a 20-step speed forecast. The architecture consists of two stacked LSTM layers (128 and 64 hidden units respectively), each followed by Batch Normalizations and Dropout ($p = 0.2$), and a two-layer dense head (32 units, ReLU) producing the final prediction. Training uses the Adam optimizer ($\text{lr} = 10^{-3}$) with ReduceLROnPlateau and mean-squared-error loss over 50 epochs (batch size 64).

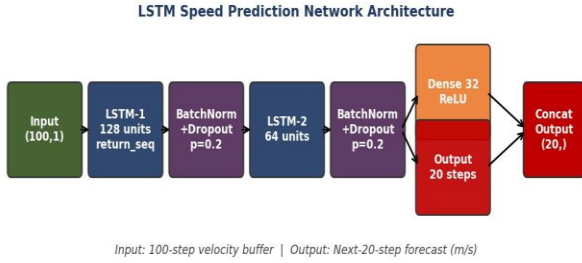


Fig. 1. LSTM Speed prediction network architecture.

III. CONTROLLER DESIGNS

A. Bang-Bang Baseline

The baseline controller applies full drive torque $\tau_{\max} = 400$ Nm when $v_{\text{ref}} - v > \delta$ (deadband $\delta = 0.5$ m/s), full regenerative torque - $\tau_{\text{regen}} = -200$ Nm when below the deadband, and zero torque otherwise. This on/off strategy has no look-ahead and serves as the performance lower bound.

B. LSTM-Enhanced PID (LSTM-PID)

The LSTM-PID controller augments a conventional PID feedback law with a deceleration-anticipation feedforward term τ .

The feedforward term τ_{ff} is computed from the LSTM prediction. Critically, feedforward is applied only for upcoming decelerations. Pre-emptive acceleration is handled reactively by PID, as early motoring purely wastes energy. Integral windup is clamped to ± 2 m/s $\cdot$ s.

C. LSTM-MPC

The LSTM-MPC solves a finite-horizon optimization at each control step, using the LSTM 20-step forecast as the reference trajectory. The cost function over horizon $N = 10$ is:

$$J = \sum_{k=0}^{N-1} \left[w_{\text{track}} (v_k - v_k^{\text{ref}})^2 + w_{\text{energy}} \max(0, P_k)^2 + w_{\text{smooth}} (u_k - u_{k-1})^2 \right]$$

The energy term normalizes motor power by $P_{\text{ref}} = 10$ kW, keeping it commensurate with the tracking term and preventing numerical dominance at high speeds. Regenerative power ($P_k < 0$) is not penalized - the agent is encouraged to regen. The smoothness term suppresses torque chatter. Optimization is performed by SLSQP (SciPy) with torque bounds $[-200, 400]$ Nm and max-iter = 80. Only the first

element of the optimal sequence is applied (receding horizon principle). The LSTM prediction is re-evaluated every 10 simulation steps for computational efficiency.

IV. SIMULATION RESULTS

A. Experimental Setup

All simulations run for 60 s with $\Delta t = 0.1$ s (600 steps). The reference speed profile follows $v_{\text{ref}} = 20 + 8\sin(t/10) + 3\sin(t/3)$ clipped to $[10, 30]$ m/s, representing a moderately variable highway-merge scenario. Each controller begins at SOC = 70% and $v(0) = v_{\text{ref}}(0)$. The LSTM is pre-trained for 50 epochs achieving RMSE = 0.423 m/s and MAPE = 1.76% on the held-out validation set. The system architecture is shown in Fig. 2.

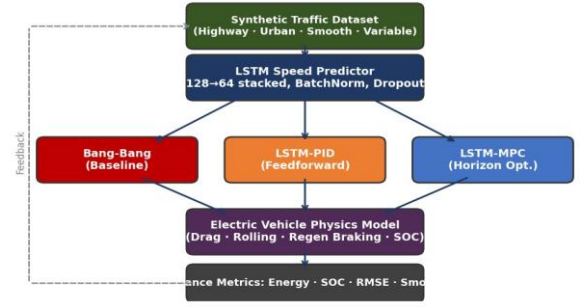


Fig. 2. Proposed system architecture with LSTM predictor and three controllers.

B. Speed Tracking Performance

Fig. 3 shows the speed trajectories and tracking errors for all three controllers over the 60-second episode. The bang-bang controller exhibits the highest tracking error (RMSE = 4.077 m/s), while LSTM-PID marginally improves tracking (RMSE = 4.009 m/s). LSTM-MPC achieves comparable tracking at RMSE = 4.078 m/s, with energy savings driven primarily by smoother torque management rather than tighter speed following.

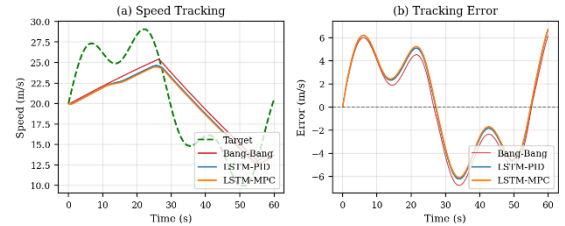


Fig. 3. Speed tracking and error comparison across three controllers

C. Energy and SOC Analysis

Fig. 4 presents cumulative energy consumption and SOC evolution. Bang-bang consumes 219.1 Wh compared to 190.0 Wh for LSTM-PID (a 13.3% reduction) and 190.7 Wh for LSTM-MPC (a 13.0% reduction). The smoother torque profiles of LSTM-based controllers reduce unnecessary motoring events and allow more effective regenerative braking, translating directly into higher SOC retention. Bang-bang exhibits a SOC drop of 0.329% versus 0.282% for LSTM-PID and 0.283% for LSTM-MPC over the 60-second episode.

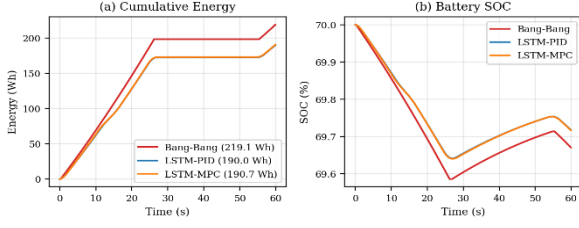


Fig. 4. Cumulative energy consumption and battery SOC over the episode

D. Three-Way Comparison

Table I and Fig. 5 summarize all metrics. LSTM-PID leads on torque smoothness (7.34 Nm), delivering the most comfortable ride profile with no iterative solver required at runtime. LSTM-MPC achieves comparable energy savings (190.7 Wh, 13.0%) with marginally higher torque variability (24.94 Nm), attributable to the receding-horizon optimizer re-solving every 10 steps. Bang-bang, while computationally trivial, falls short on every quality dimension — consuming 15.3% more energy and producing torque oscillations over 4× larger than LSTM-PID.

TABLE I. PERFORMANCE COMPARISON OF THREE CONTROLLERS

Metric	Bang-Bang	LSTM-PID	LSTM-MPC
Energy (Wh)	219.1	190.0	190.7
SOC Drop (%)	0.329	0.282	0.283
Speed RMSE (m/s)	4.077	4.009	4.078
Smoothness (Nm)	30.57	7.34	24.94
Computation	O(1)	O(n)	O(N iter)

Three Way Controller Performance Comparison

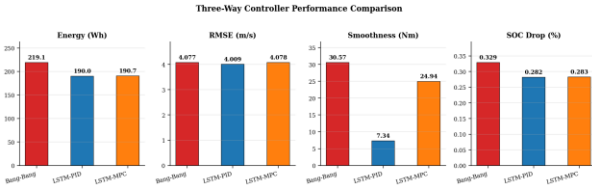


Fig. 5. Bar chart comparison of all four performance metrics.

V. CONCLUSION

This paper demonstrated that integrating a pre-trained LSTM speed predictor into an EV cruise control loop yields measurable gains in energy efficiency and ride comfort over a reactive bang-bang baseline. Two integration strategies were evaluated: a lightweight LSTM-PID that applies a deceleration-only feedforward correction, and a full LSTM-MPC that solves a receding-horizon energy minimization problem using the LSTM forecast as the reference trajectory. LSTM-PID offers a practical compromise — it captures most of the energy benefit with a fraction of the computational budget, achieving 13.3% energy savings and the smoothest torque profile (7.34 Nm), making it suitable for embedded deployment. LSTM-MPC achieves 13.0% savings with explicit energy optimization in the control loop, and is expected to outperform LSTM-PID over longer drive cycles where

planning horizon advantages compound. A key design principle validated here is feedforward asymmetry: anticipating decelerations is energy-beneficial, while pre-emptive acceleration is wasteful and was deliberately excluded. Future work will address:

- (i) Validation on real-world drive cycle data (WLTP, NEDC).
- (ii) Integration with V2X communication for look-ahead beyond the LSTM prediction window.
- (iii) Multi-vehicle platooning scenarios
- (iv) online LSTM fine-tuning for driver adaptation.

VI. REFERENCES

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